NETWORK WALKS

Cybersecurity Internship Program

Virtual Lab Network Configuration Report

NAT Network setup for Kali Linux and Windows XP in VirtualBox

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Date	29 October 2026

Part A: VirtualBox NAT Network

Figure 1: Opening the VirtualBox Network Manager

In Oracle VirtualBox Manager, the File menu was opened and Tools > Network was selected. This opens the Network Manager, where NAT Networks, Host-only Networks and Cloud Networks are created and managed.

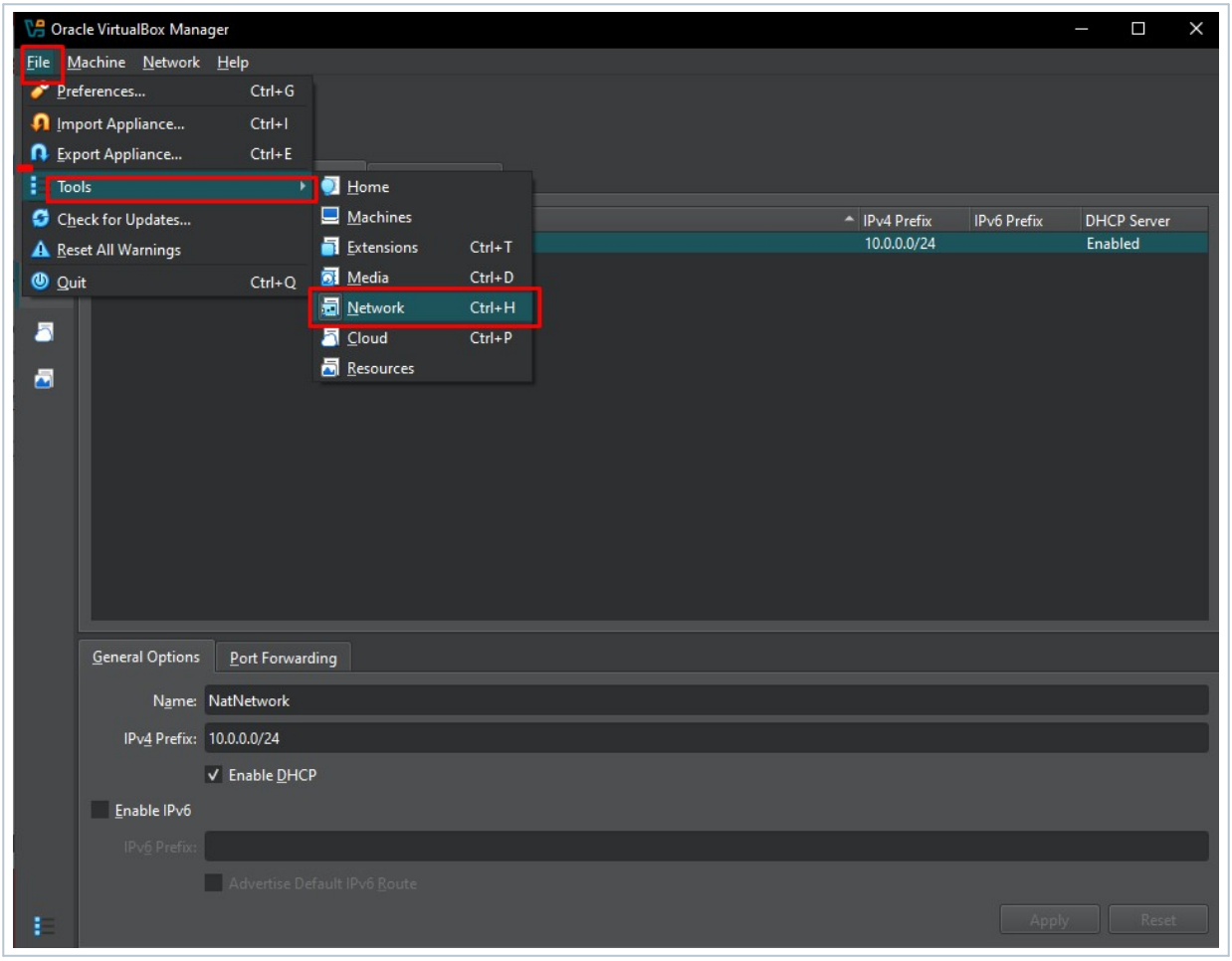
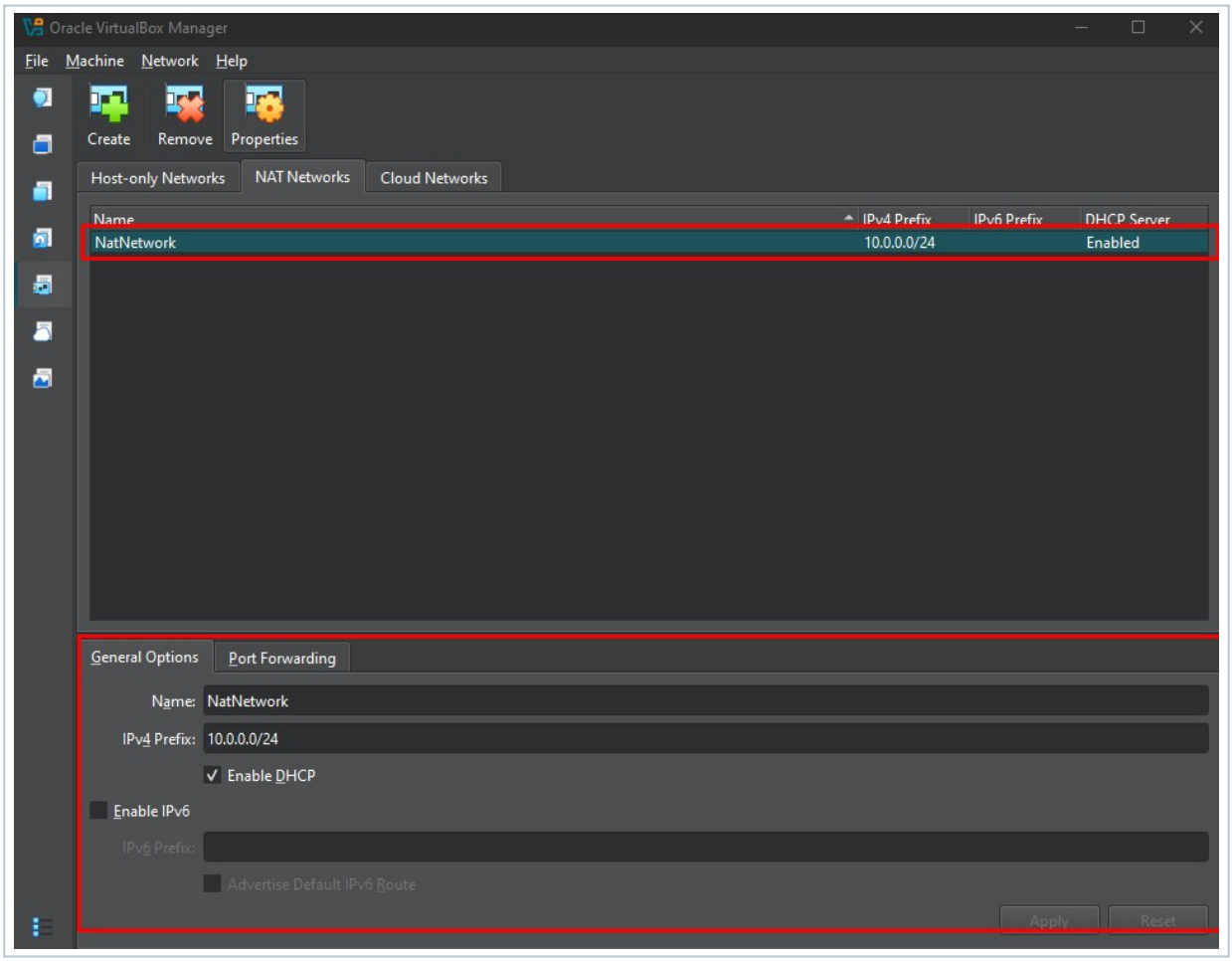


Figure 2: NAT Network Configuration

A NAT Network named **NatNetwork** is configured with the IPv4 prefix **10.0.0.0/24** and DHCP enabled. Every virtual machine attached to it shares this subnet and can communicate with the other VMs and the internet.



Part B: Software and Virtual Machines

Figure 3: Downloaded VM Images and Setup Files

The lab files were downloaded and stored on the local disk (D:), including the Kali Linux VirtualBox image and the Windows XP image. The required archives are highlighted in the folder.

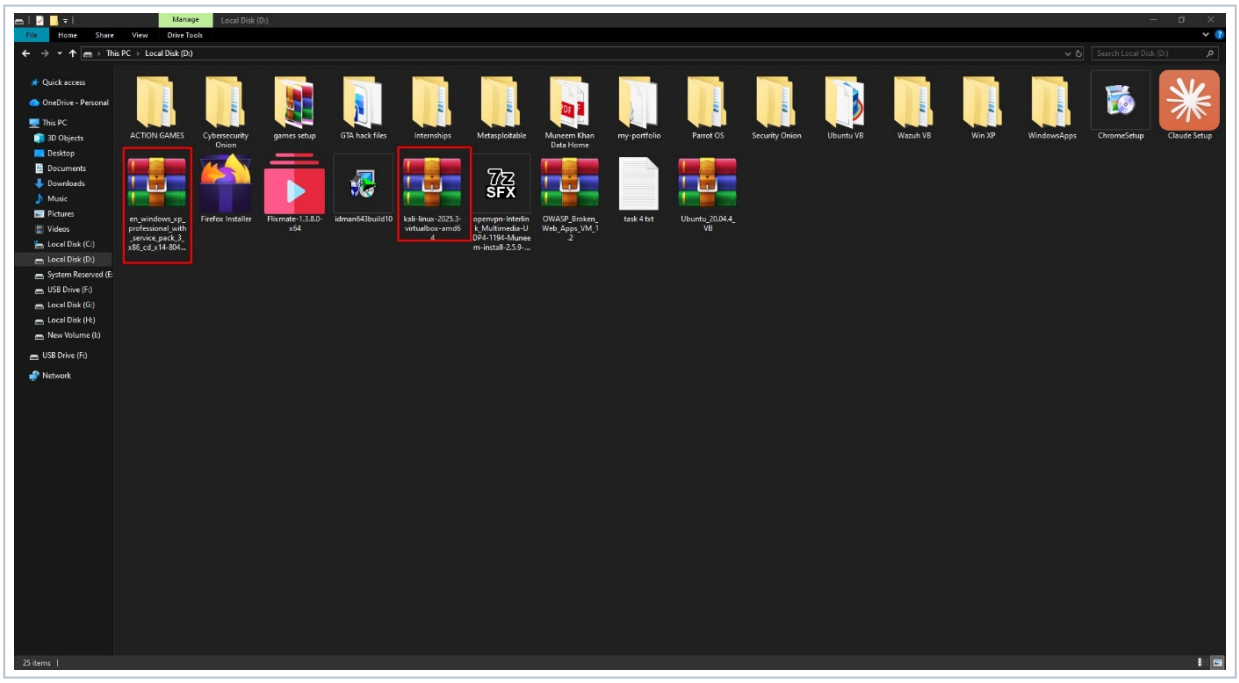


Figure 4: Virtualization Software Installers

The installers for **VirtualBox 7.2.6a** and **VMware Workstation Pro 17.6.2** were prepared as the virtualization platforms for the lab, along with the VirtualBox Extension Pack.

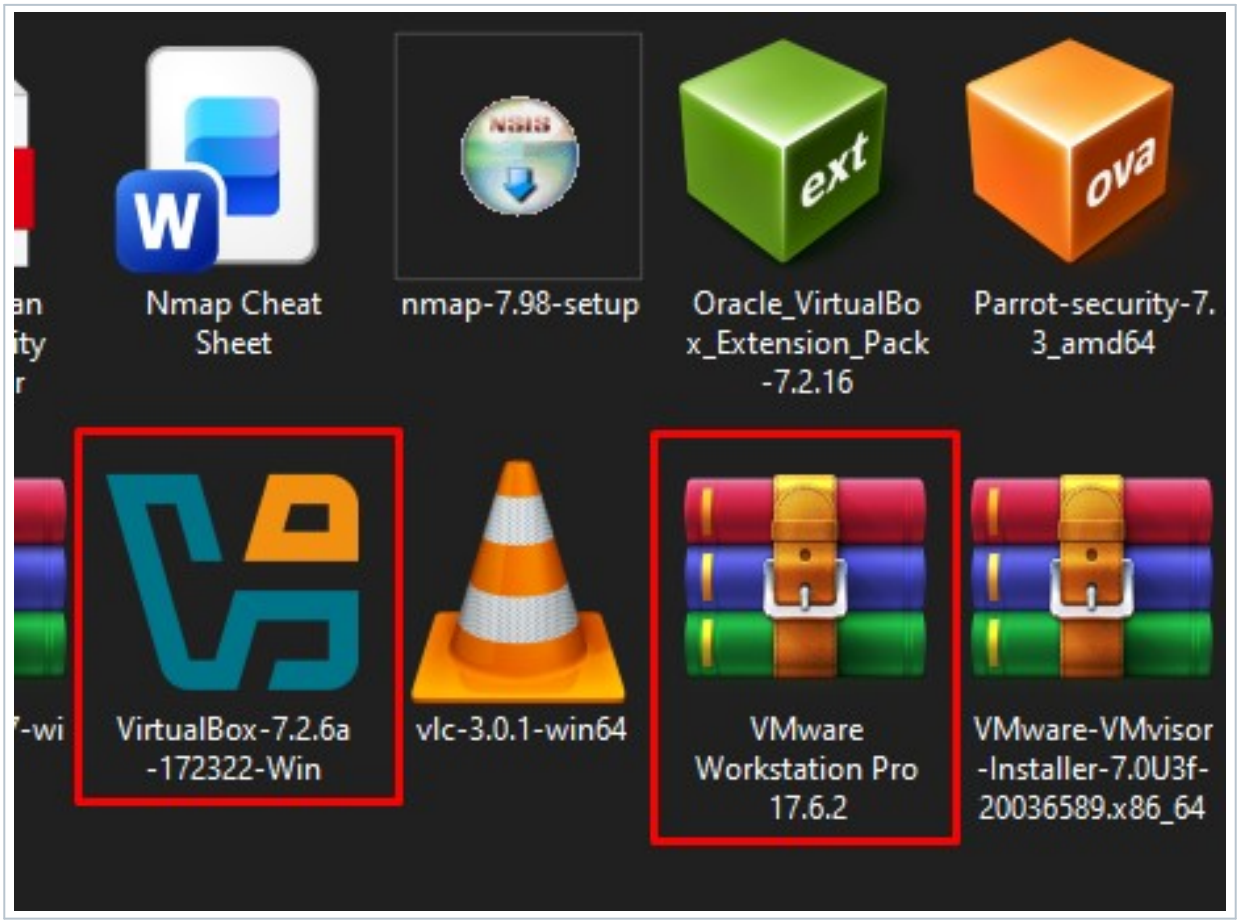
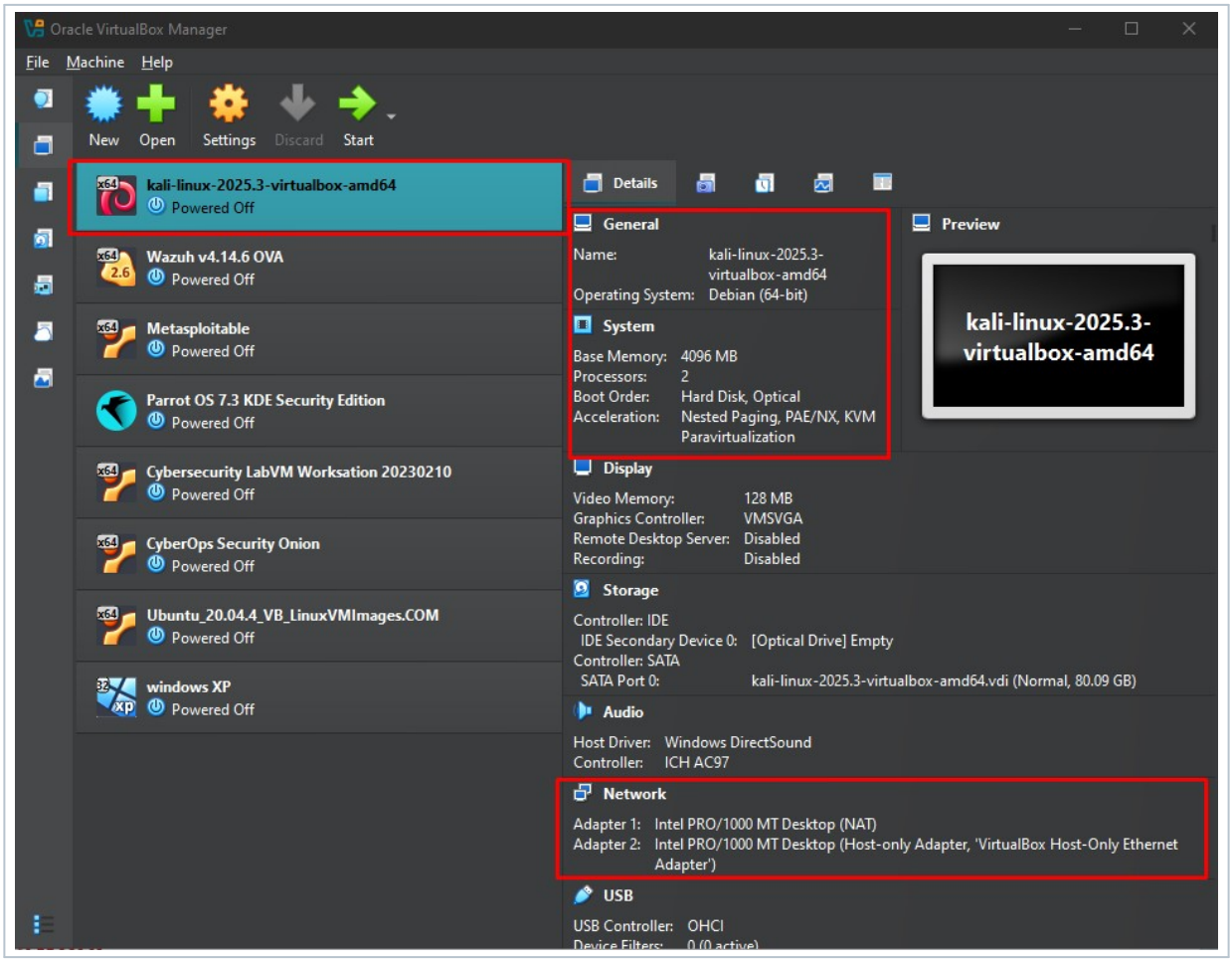


Figure 5: Kali Linux VM Overview

The Kali Linux 2025.3 virtual machine was imported into VirtualBox. Its details show 4096 MB of memory, 2 processors, and two network adapters: Adapter 1 (NAT) and Adapter 2 (Host-only).



Part C: Network Adapter Settings

Figure 6: Adapter 1: Default NAT Mode

Adapter 1 of the Kali VM was initially attached to **NAT** using the Intel PRO/1000 MT Desktop adapter. In this mode the VM can reach the internet but is isolated from other virtual machines.

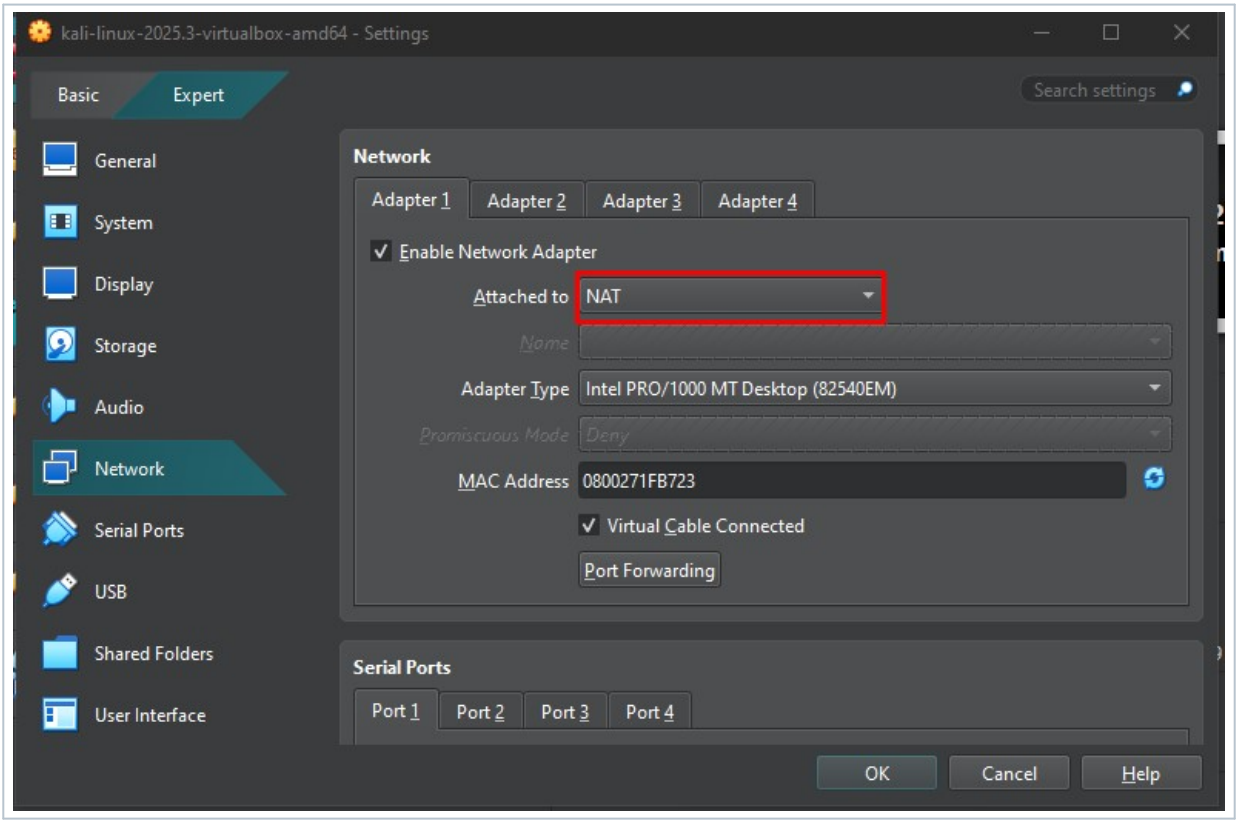


Figure 7: Adapter 2: Host-only Adapter

Adapter 2 is attached to a **Host-only Adapter** (VirtualBox Host-Only Ethernet Adapter). This provides a private link between the Kali VM and the host machine.

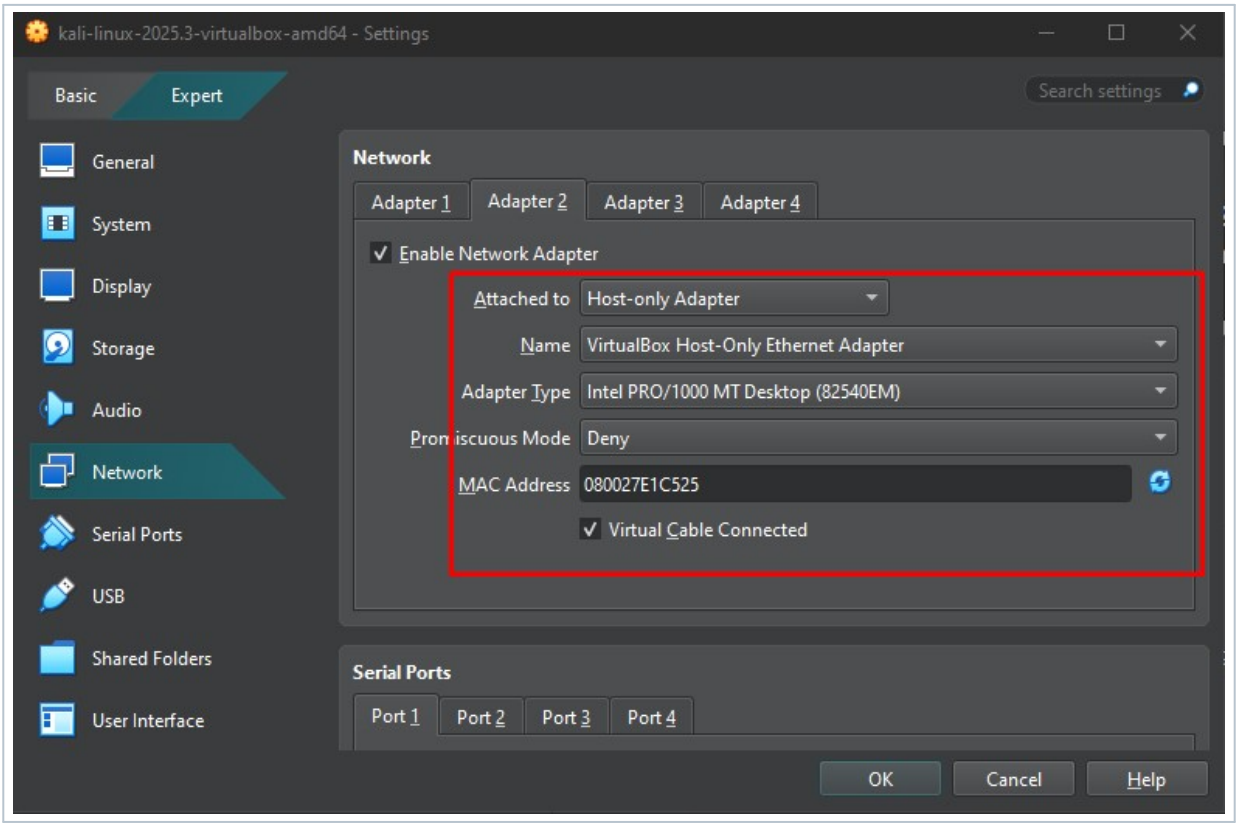
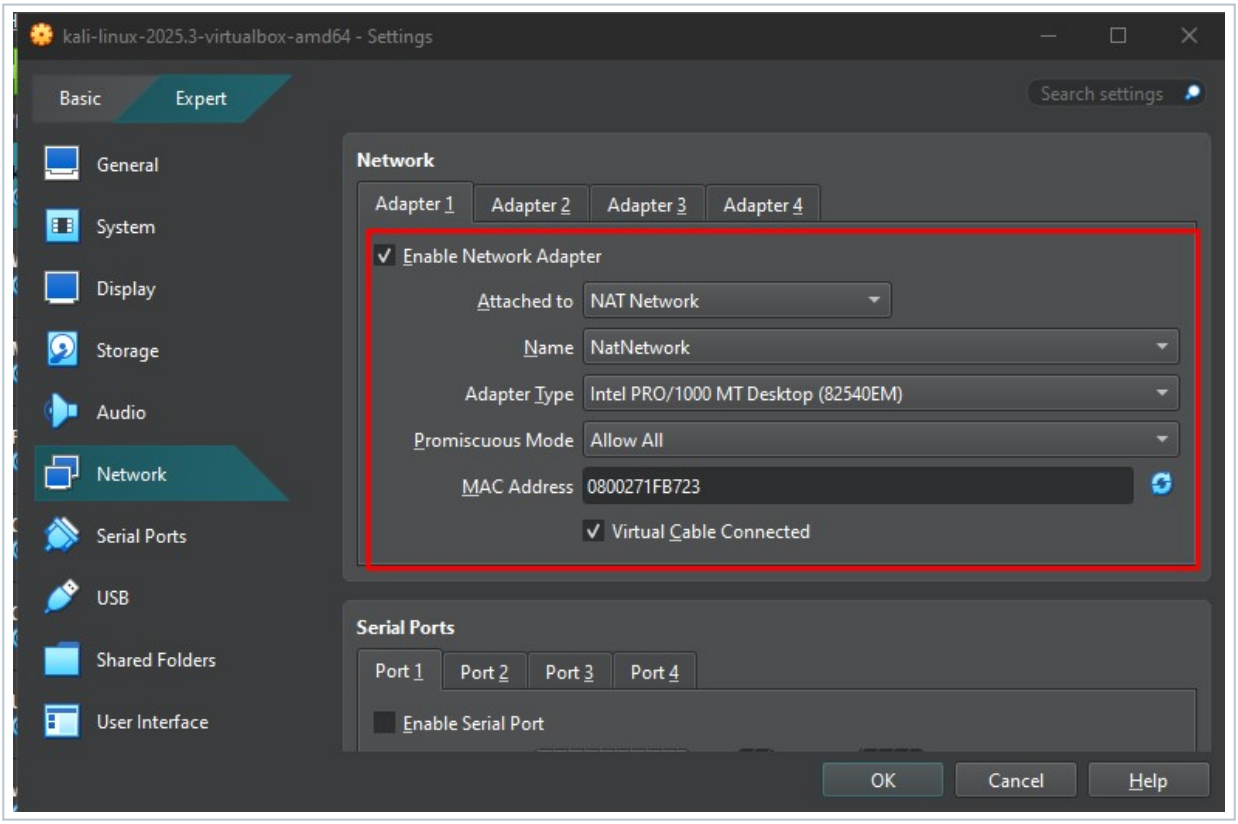


Figure 8: Adapter 1 Changed to NAT Network

Adapter 1 was switched from NAT to **NAT Network** and the network **NatNetwork** was selected. Promiscuous Mode was set to **Allow All** so the VM can capture and analyze traffic on the virtual network.



Part D: IP Configuration in Kali Linux

Figure 9: Opening Network Connections in Kali

Inside the running Kali VM, the network icon in the top panel was used to open **Edit Connections**. The Network Connections window lists the Ethernet profiles Wired connection 1 and Wired connection 2.

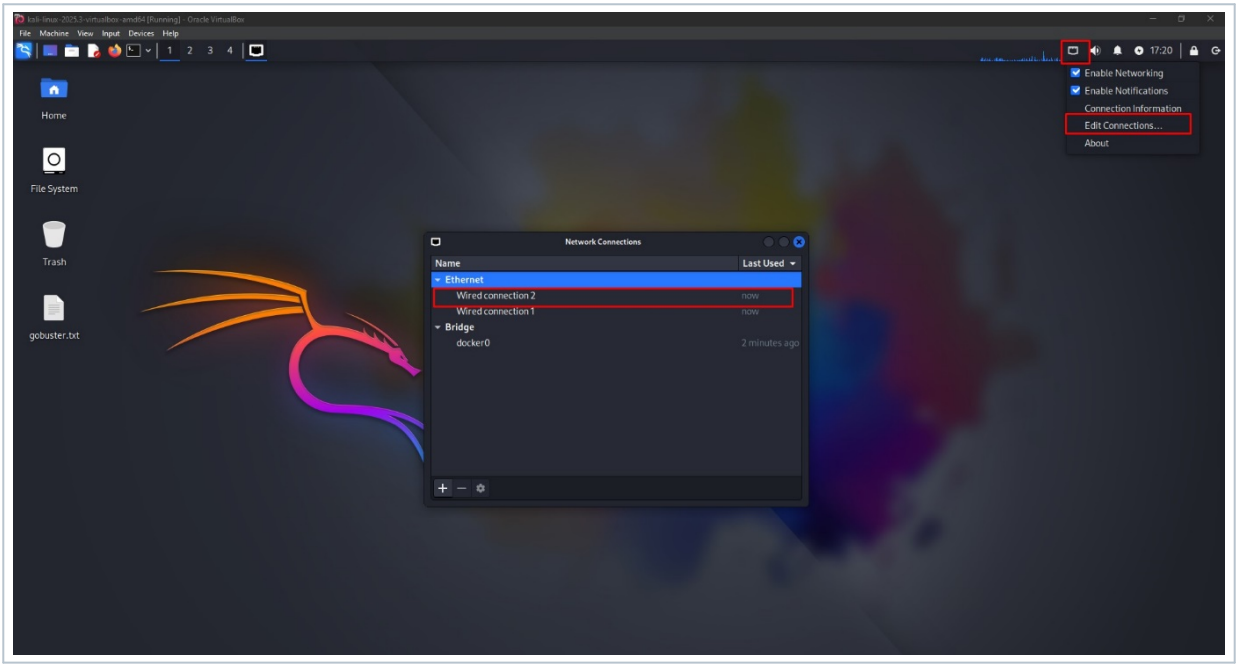


Figure 10: Manual IPv4 Settings

In the IPv4 Settings tab of Wired connection 1, the method was set to **Manual** with address **10.0.0.2**, netmask **24** and gateway **10.0.0.1**. The DNS server **8.8.8.8** was added to fix internet connectivity issues.

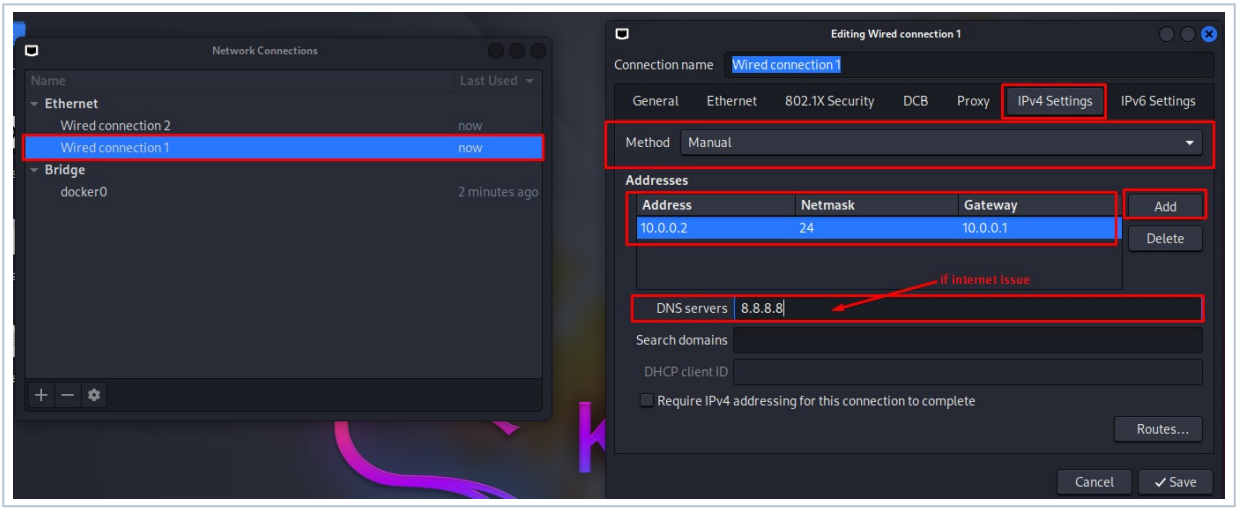


Figure 11: Verifying the IP Address with ip addr

The `ip addr` command was run to verify the interface configuration. The eth0 interface shows an address in the **10.0.0.0/24** network (10.0.0.3/24), confirming that Kali is connected to NatNetwork.

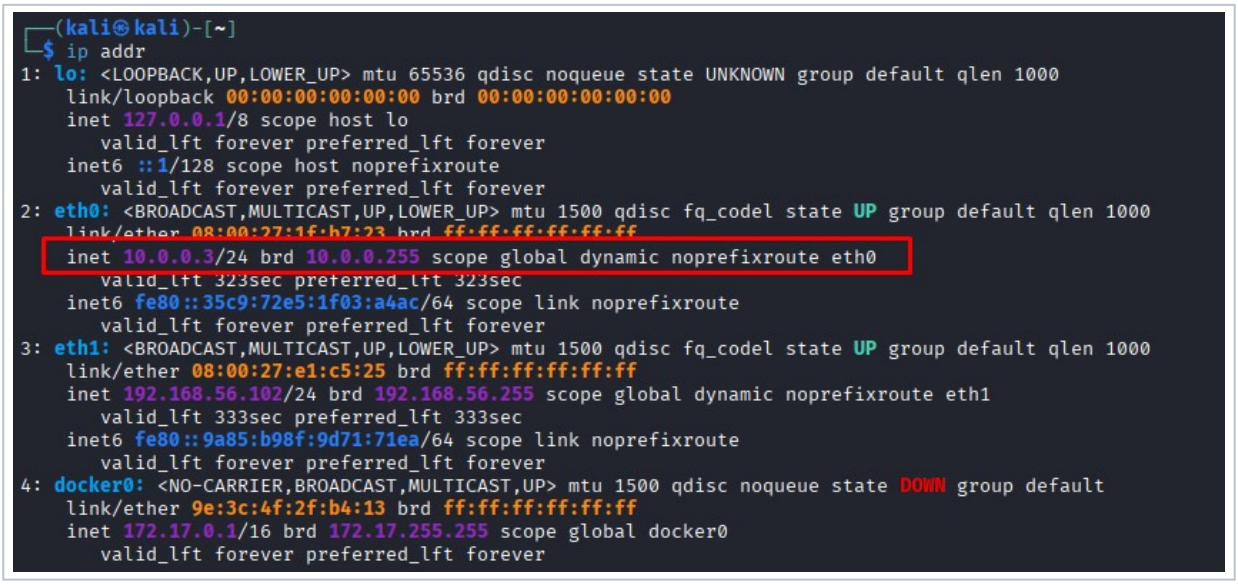


Figure 12: Setting the DAD Timeout to 0

The `nmcli` command was used to set `ipv4.dad-timeout` to 0 for Wired connection 1. This removes the duplicate address detection delay so the interface becomes active faster.

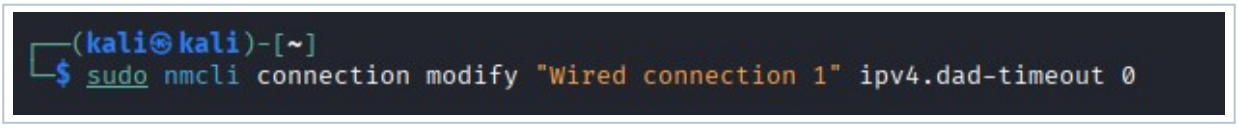


Figure 13: Reactivating the Connection

The connection was brought up again with `nmcli connection up`. NetworkManager reported a successful activation, which applies the new settings.

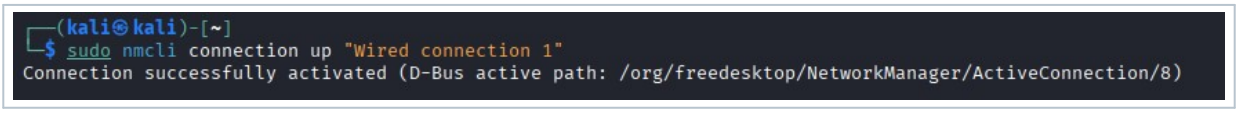


Figure 14: Verifying the DAD Timeout Value

The command `nmcli -g ipv4.dad-timeout connection show` returned **0**, confirming that the setting was saved correctly.



Part E: Kali Linux VM Settings

Figure 15: System Settings: Memory and Boot Order

In the System settings of the Kali VM, base memory is set to **4096 MB** and the boot order is Hard Disk followed by Optical. I/O APIC is enabled and the pointing device is set to USB Tablet.

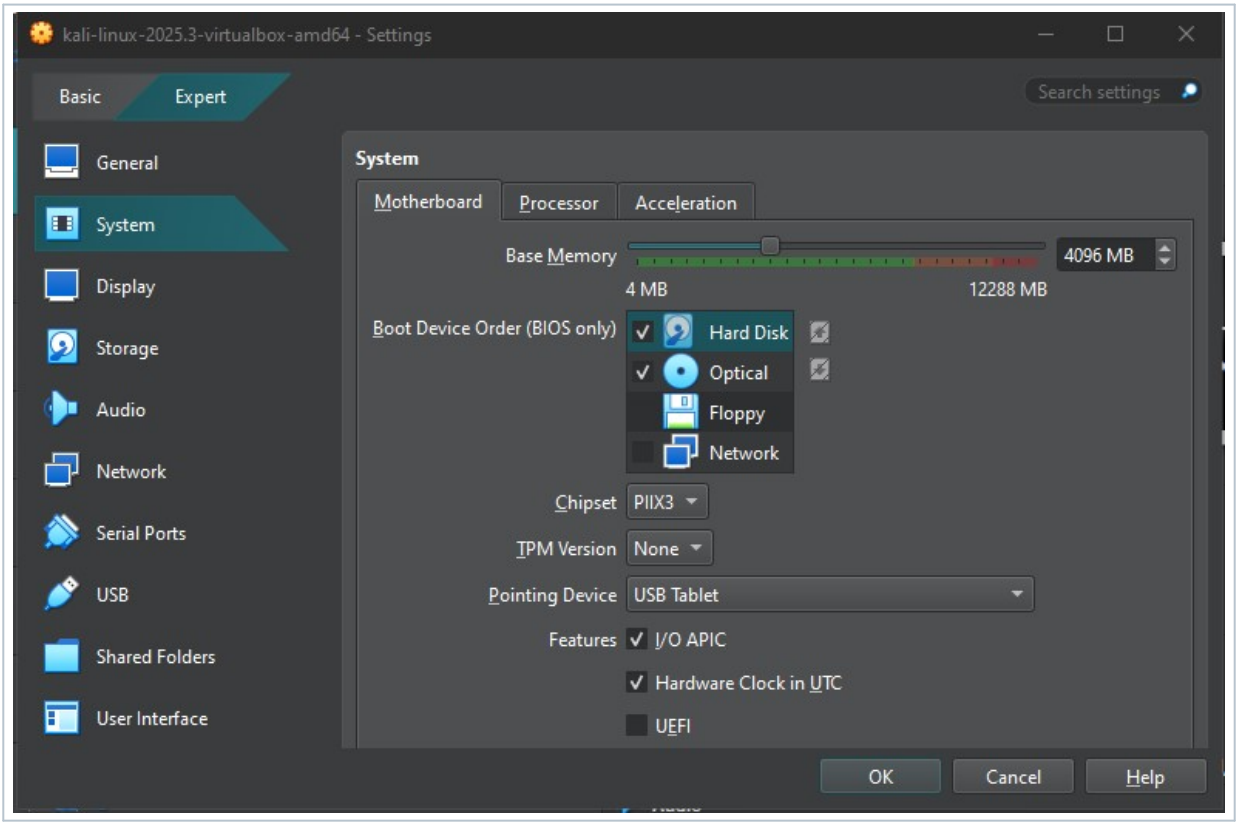


Figure 16: System Settings: Acceleration

Under the Acceleration tab, the paravirtualization interface is set to Default and hardware virtualization with **Nested Paging** is enabled to improve VM performance.

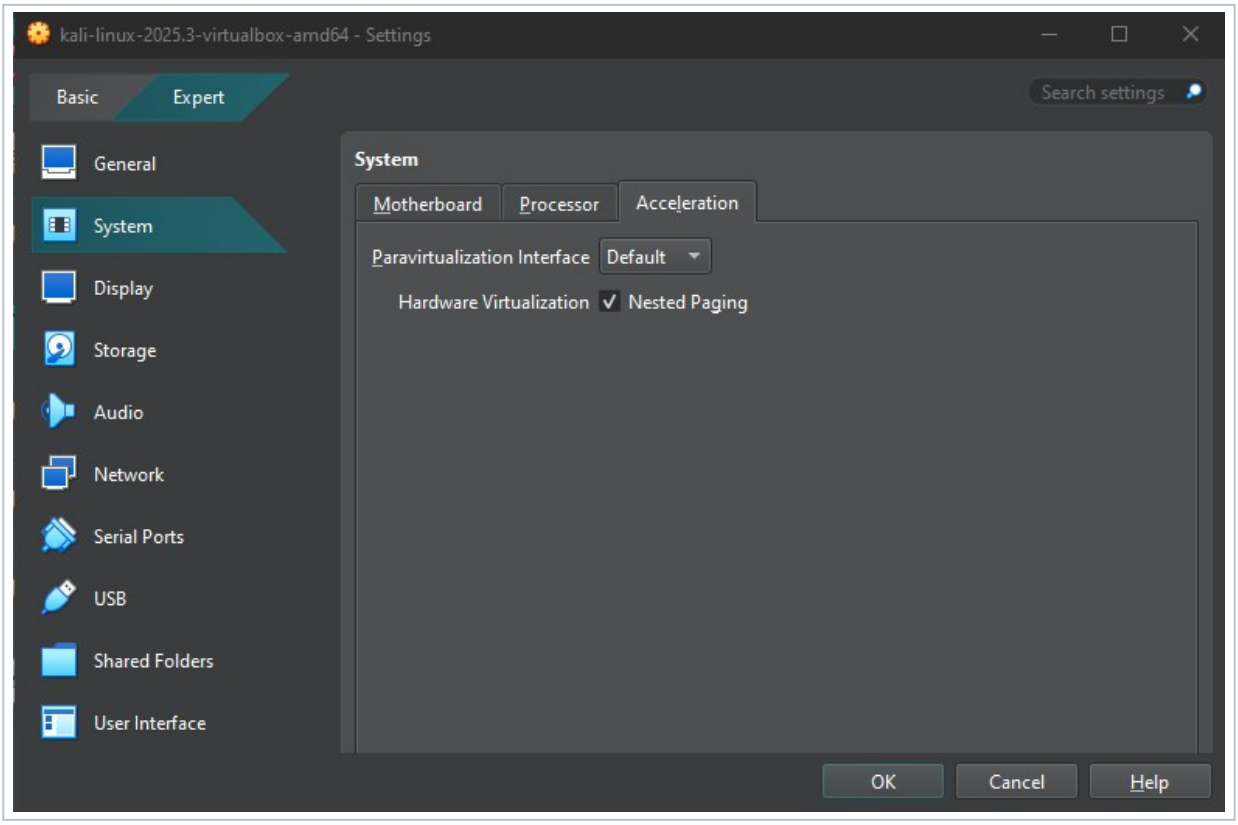
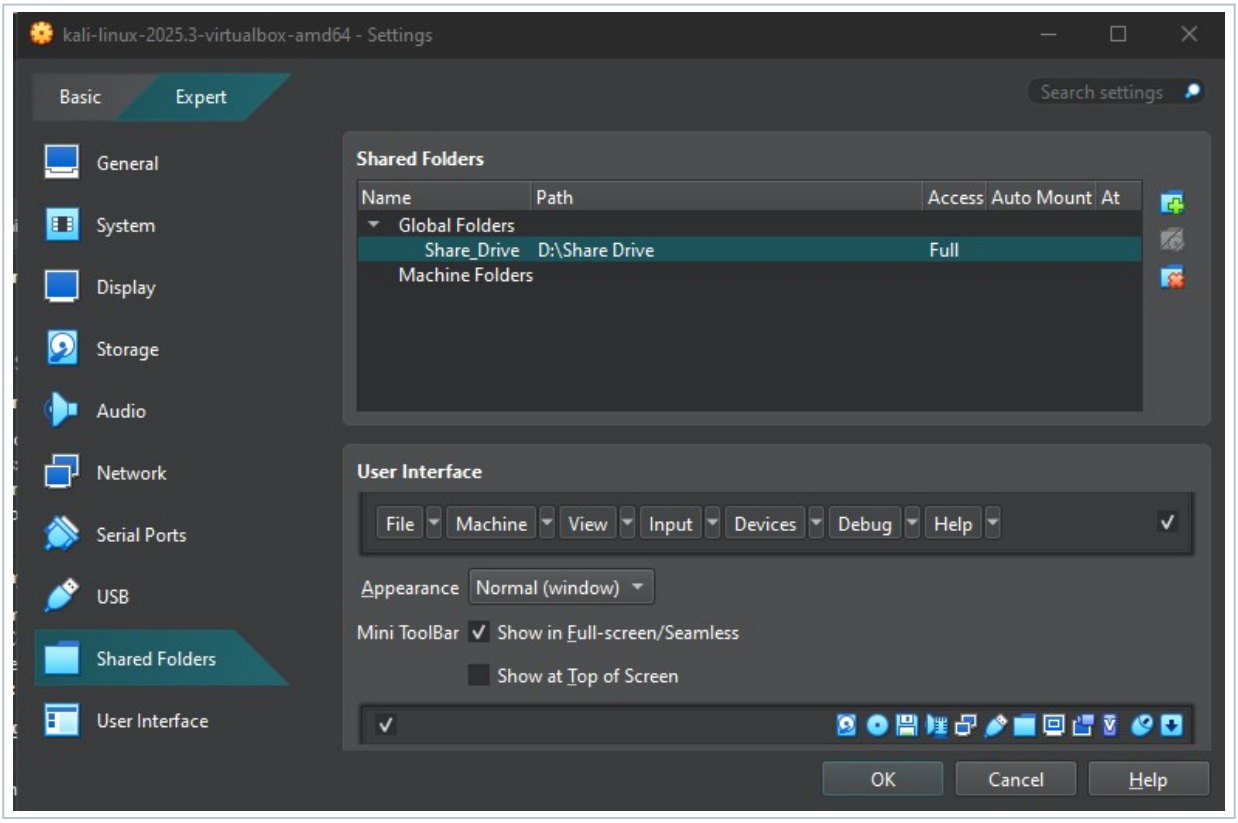


Figure 17: Shared Folder and User Interface Settings

A global shared folder named **Share_Drive** pointing to D:\Share Drive was added with Full access, allowing file transfer between the host and the VMs. The User Interface tab controls the menu bar and mini toolbar.



Part F: Windows XP Virtual Machine

Figure 18: Windows XP Network Configuration

The Windows XP VM is configured the same way: Adapter 1 is attached to **NAT Network (NatNetwork)** with Promiscuous Mode set to Allow All. This places it on the same 10.0.0.0/24 network as the Kali VM so the two machines can communicate.

